

Student Performance Dataset – Report for Graphs (Data Visualization)

1. The boxplots show that there are some differences between male and female students in their math and reading scores. Female students have a higher median reading score, while male students have a slightly higher median math score. Both sets of scores overlap between the two groups, so there is not a major difference overall. There are also several lower-score outliers for both male and female students, with more lower outliers appearing in the female math scores. Overall, the graph suggests that there are some gender differences that are more noticeable in the reading scores. It also suggests that there are some differences in math scores, but the difference is smaller.
2. The bar graph shows that students who completed the test preparation course had higher average math scores than students who did not complete the course. Students who completed the test preparation course had an average math score of about 69 points, while students who did not complete it had an average score of about 64 points. This is a difference of about 5 points between the two groups. Although the difference is not very large, the graph shows a noticeable difference in the average math scores. This suggests there may be a relationship between completing the test preparation course and math performance. Overall, students who completed the test preparation course had higher average math scores.
3. The bar graph shows that lunch type is related to differences in students' overall performance. Students with a standard lunch had a mean overall score of about 71 points, while students with a free/reduced lunch had a mean overall score of about 62 points. This is a difference of about 9 points between the two groups. The difference is noticeable and shows that students with a standard lunch had higher scores overall. This suggests that there may be a relationship between lunch type and students' overall academic performance. Overall, students with a standard lunch had higher average scores than students with a free/reduced lunch.
4. The heatmap shows that math, reading, and writing scores all have strong positive correlations, meaning the three subjects tend to move together. The math and reading scores have a correlation of 0.82, while the math and writing scores have a correlation of 0.80. The reading and writing scores have the strongest correlation at 0.95. This shows that students who score higher in one subject also tend to score higher in the other subjects. The strongest relationship is between the reading and writing scores because their correlation is the closest to 1. The results also show that none of the subjects have a negative relationship with one another. Overall, all three subjects have strong positive relationships, with the reading and writing scores moving together the most closely.
5. The scatter plot shows a strong positive relationship between the math and reading scores for both test preparation groups. Students with higher reading scores generally also have higher math scores, which is shown by the upward trend in the data. The trend lines for students who completed the test preparation course and those who did not both move upward and have fairly similar slopes. This means that math scores tend to increase as reading scores increase at a similar rate for both groups. There does not appear to be a large difference in the slopes based on test preparation status. Overall, math and reading scores are strongly associated, and the relationship between the two scores is similar for students who completed the test preparation course and those who did not.